ISTANBUL UNIVERSITY-CERRAHPAŞA
Department of Computer Engineering
BIMU2059-DIFFERENTIAL EQUATIONS
Midterm (Weight, 50%)

Student's Name	
Student's ID Number	

Academic Year	2025-2026	**Date of Exam**	24.11.2025 (15:20)
Professor	Assoc. Prof. Dr. Uğur Odabaşı	**Time Duration (min)**	**65**

Question#	**PQ\LO**	**Grade**	**Question#**	**PQ\LO**	**Grade**
1.	PQ: 1 \ LO: 1, 2, 3	/25	**3.**	PQ: 1 \ LO: 1, 2, 3	/25
2.	PQ: 1 \ LO: 1, 2, 3	/25	**4.**	PQ: 1 \ LO: 1, 2, 4	/25

Total /100

QUESTIONS

1. $$(\sqrt{x^2 - y^2} + y)\,dx - x\,dy = 0$$

   a) Find the general solution.

   b) Find any particular solution not obtainable from the general solution (singular solution).

(15 + 10 = 25 Pts)

2. Solve the initial value problem:

$$\frac{dy}{dx} = \frac{xy^2 - \cos x \sin x}{y(1 - x^2)}, \quad y(0) = 2$$

(25 Pts)

3. a) Prove that every separable first-order differential equation is exact.

   b) Given first order nonlinear differential equation

$$y' + \tfrac{7y}{x} - 3y^2 = \tfrac{3}{x^2}$$

   i) Prove that after substitution, $y(x) = \frac{1}{x} + u(x)$, the given equation can be transformed into a Bernoulli differential equation;

   ii) Solve the resulting Bernoulli differential equation.

(10 + 2 × 7.5 = 25 Pts)

4. Verify that the functions $\{1, x, \cos x, \sin x\}$ form a fundamental set of solutions of the differential equation

$$y^{(4)} + y'' = 0$$

on the interval $(-\infty, \infty)$, and form the general solution.

(25 Pts)